Student Notebook

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Welcome to Elite Aviation's Private Pilot Program!

You'll find that the Syllabus guide below is a tool to guide you through our course. It provides a syllabus overview, lesson objectives, homework assignments, resource list, and questions to answer or ask your instructor.

For the best results through the course, we recommend following the guide as intended and coming in prepared for each lesson. This includes completing the homework assignment and answering the study questions on your own. The answer to each study question can be found in the homework or resource material. If you find a particular question or topic to be difficult, don't sweat it! That is why we have a team of highly qualified Certified Flight Instructors to help you.

When you go over the material, keep in mind the four building blocks to successful learning.

- **Rote**: Rote knowledge is knowledge that is purely memorized. Often you do not need context for this knowledge. You'll experience Rote Knowledge most in the beginning of your training. Such as learning of the speeds at which your aircraft operates.
- **Understanding:** Understanding occurs when you take the Rote Knowledge and apply context to it. You'll see this the most during the beginning of your training. An example of this is when you understand the reason your aircraft has certain operating speeds for different maneuvers.
- **Application**: Application occurs when you take both Rote and Understanding and use it in a real-world example. Such as when you memorize the steps of a maneuver, understand what occurs aerodynamically during that maneuver, and then be able to perform that maneuver successfully. You'll experience this level of learning throughout your training.
- **Correlation:** Correlation is the highest level of learning. It is when you take two separate concepts and find a correlation between the two, thus enhancing your experience. This level of learning will take place towards the end of your training, once you understand most major concepts and can apply them.

By understanding which level of learning you're working towards, you can most effectively plan your progress forward. If you have trouble determining what level of learning to work on, ask your assigned instructor!

Again, we welcome you to the program and we are excited to assist you in your journey in becoming a pilot. Here at Elite Aviation, we pride ourselves on our quality, expertise, and customer experience. If you have any questions at all we would be pleased to help you.

Congratulations in getting started,

Allison Lewandowski
Flight Operations Manager|Elite Aviation

Syllabus Overview –Track your progress				
Note: The "Time" column only represents the minimum flight time or the maximum ground time. It does not include the pre & post brief time. These are only estimates and further instruction may be needed at any given time to bring you as the student up to a proficient level.				
Lesson Number	Activity	Title	Date Completed	Time
Phase 1 – Pre-solo				
Lesson 1	Ground	Intro to Elite Aviation & Risk Management		2
Lesson 2	Flight	Preflight & Four Fundamentals		1.2
Lesson 3	Ground	Basic Aircraft Systems & Limitation		2
Lesson 4	Flight	Preflight & Four Fundamentals		1.2
Lesson 5	Ground	Four Forces & Slow Flight		2
Lesson 6	Flight	Preflight & Four Fundamentals		1.2
Lesson 7	Ground	Aerodynamics, Stalls & Steep Turns		2
Lesson 8	Flight	Four Fundamentals & Slow Flight		1.2
Lesson 9	Ground	Airport Ops & Radio Calls		2
Lesson 10	Flight	Maneuvering in Slow Flight		1.2
Lesson 11	Flight	Slow Flight, Stalls, & Steep Turns		1.2
Lesson 12	Flight	Maneuver Review		1.2
Lesson 13	Ground	Ground Reference Maneuvers		2
Lesson 14	Flight	Ground Reference Maneuvers		1.2
Lesson 15	Flight	Traffic Pattern & Landing		1.2
Lesson 16	Ground	Emergencies		2
Lesson 17	Flight	Emergencies		1.2
Lesson 18	Flight	Emergencies 2		1.2
Lesson 19	Flight	Traffic Pattern & Landings		1.2
Lesson 20	Ground	Aircraft Systems		2
Lesson 21	Flight	Traffic Pattern & Landings		1.2
Lesson 22	Ground	Airspace & Aeronautical Charts 1		2
Lesson 23	Flight	Traffic Pattern & Landings		1.2
Lesson 24	Ground	Weight & Balance, Performance		2
Lesson 25	Flight	Traffic Pattern & Landings		1.2
Lesson 26	Ground	Aircraft Inspections, Inop Equip, Docs.		2
Lesson 27	Flight	Traffic Pattern & Landings		1.2
Lesson 28	Flight	Review		1.2
Lesson 29	Ground	Oral Review		2
Lesson 30	Ground	Pre-solo Prog Check		2
Lesson 31	Flight	Pre-solo Prog Check		1.2
Lesson 32	Flight	Pattern & First Solo		1.2

Phase 2 – Pre-Solo Cross Country				
Lesson 33	Ground	Navlogs & Aeronautical Charts 2		2
Lesson 34	Ground	Short & Soft Field Operations		2
Lesson 35	Flight	Short & Soft Field Operations		1.2
Lesson 36	Flight	Cross Country Operations 1		1.5
Lesson 37	Ground	Weather Reporting Systems & Briefs		2
Lesson 38	Flight	Cross Country Operations 2		1.5
Lesson 39	Ground	Weather Theory		2
Lesson 40	Flight	Cross Country Operations 3		1.5
Lesson 41	Ground	BAI & Unusual Attitudes		2
Lesson 42	Ground	Night Ops & Aeromedical Factors		2
Lesson 43	Flight	Night Takeoffs & Landings		1.5
Lesson 44	Flight	Night Cross Country		1.5
Lesson 45	Sim	Lost Procedures, VOR & GPS Operation		2
Lesson 46	Flight	Lost Procedures, Diversion		1.5
Lesson 47	Flight	Cross Country Ops/Diversions		1.5
Written must be passed to continue past this point				
Lesson 48	Ground	Pre-solo XC Oral Prog Check		1.5
Lesson 49	Flight	Pre-solo XC Prog Check		1.5
Phase 3 –Checkride Prep				
Lesson 50	Flight	First Solo XC		1.5
Lesson 51	Flight	Second Solo XC		1.5
Lesson 52	Flight	Long Solo XC		2
Lesson 53	Flight	Dual Maneuvers Evaluation		1.5
Lesson 54	Ground	Private Pilot Privileges & Limitations		2
Lesson 55	Flight	Solo to Practice Area 1		1.5
Lesson 56	Flight	Solo to Practice Area 2		1.5
Lesson 57	Flight	Solo to Practice Area 3		1.5
Lesson 58	Ground	Mock Oral		2
Lesson 59	Flight	Mock Checkride		1.5
Lesson 60	Ground	Logbook Review		2
Lesson 61	Ground	End of Course		2
Lesson 62	Flight	End of Course		1.5
End of Course must be passed satisfactorily to move on to checkride				

Lesson 1, Ground 1: Intro to Elite Aviation & Risk Management

Date Completed: Instructor Signature:

Lesson Objective: To understand the Flight Risk Analysis Tool (FRAT) sheet, be able to decode a METAR, and be able to identify the basic structures of the airplane. V-speeds will also be introduced during this lesson. Homework must be completed before the lesson.

Resources for lesson:

FAR 61.83, 61.89; Student Pilot Requirements & Limitations
FRAT Sheet
METAR Decoder
Pilot's Operating Handbook (Cessna 172 S model Nav 2) Chapter 2. You can find this is your Cessna Training Application

Homework:

Rote Knowledge: V-Speed Worksheet. Using the POH, fill out the associated V-speeds. All speeds are found in chapters 1 & 2. The goal of this homework is to 1) familiarize yourself with the POH and 2) commit the V-speeds to memory.

Fill out a practice FRAT sheet
CTA Stage 1, Phase 1, Section 1

What is a V-Speed?_____

Lesson 2, Flight 2: Preflight and Four Fundamentals

Date Completed: Instructor Signature:

Lesson Objective: To review the FRAT sheet, learn why and how we preflight, and what a "sight picture" is. The lesson will include basic maneuvering such as climbs, turns, descents, and straight and level flight. Checklist usage will also be emphasized. The student may conduct the takeoff with guidance from the instructor. Homework must be completed before the lesson.

Homework:

CTA Stage 1, Phase 1, Section 2
CTA Stage 1, Phase 3, Section 2
V-Speeds from memory

What is "rotation" and what speed do we perform it?_____

At what speed do we climb out?_____

Lesson 3, Ground 2: Basic Aircraft Systems & Limitations

Date Completed: Instructor Signature:

Lesson Objective: To identify which instruments are pitot static and which are vacuum systems. Using that knowledge and be able to translate the information from those instruments. An introduction of flight controls and aircraft limitations will be included. Finally, a review of V-speeds from memory will be conducted. Homework must be completed before the lesson.

Homework:

PHAK Ch. 3 Construction
PHAK Ch. 6 Flight Controls
FAR 61.83,61.89
CTA Stage 1, Phase 1, 3

Which instruments are Pitot-Static?

_____

Which instruments are Vacuum powered? _____

What are the four cardinal headings on a compass? _____

What are the primary control surfaces? _____

What are the secondary control surfaces? _____

Lesson 4, Flights 3: Preflight & Four Fundamentals 2

Date Completed: Instructor Signature:

Lesson Objective: To review all topics covered in Flight 1 to reaffirm the sight picture. Checklist usage will again be emphasized. Student shall perform the takeoff. Homework must be completed before the lesson.

Homework:

CTA Stage 1, Phase 1, 4

What is the purpose of the three-way exchange of controls?_____

_____

_____

_____

What is included in the passenger briefing?

_____

_____

Lesson 5, Ground 3: Four Forces of Flight & Slow Flight

Date Completed: Instructor Signature:

Lesson Objective: To introduce the students to the basics of lift creation, drag, weight and thrust, how the angle of attack relates to pitch and airspeed, the aerodynamics of flaps, wake turbulence. The student must be able to understand how these principles apply to flight. Forces in a turn, adverse yaw, and rate/radius will be introduced. V-speeds will be reviewed by testing the student's memory. Homework must be completed before the lesson.

Homework:

PHAK Ch. 4 Principles of Flight
Elite Aviation's Maneuver Sheet – Slow Flight
Memory Items Sheet –Slow Flight (pg. 50, memorize these steps)
CTA Stage 1, Phase 2, Section 4

How is lift created? Draw and/or explain how it occurs.

What are the types of drag?

_____

What are wingtip vortices? Draw and explain how they occur.

What are the dangers of wingtip vortices?

_____

_____

What are the four left turning tendencies? Explain how they work.

1_____

_____

2_____

_____

3_____

_____

4_____

_____

Lesson 6, Flight 4: Preflight & Four Fundamentals 3

Date Completed: Instructor Signature:

Lesson Objective: To review sight picture and four fundamentals. Landing will be demonstrated and discussed. Student shall "trace" the instructor on the controls.

Homework:

CTA Stage 1, Phase 3, Section 3

What are the steps to slow flight?_____

_____

_____

_____

Lesson 7, Ground 4: Aerodynamics, Stalls, & Steep Turns

Date Completed: Instructor Signature:

Lesson Objective: To review the four forces of flight. It's important for the student to fully understand the concept of an aerodynamic stall and spin with associated recovery procedures. Also, the student should strive to memorize maneuver procedures and "chair fly". Chair flying is the act of sitting in the plane while acting out the procedure.

Homework:

Elite Aviation's Maneuver Sheet –Power On & Power Off Stalls
Memory Items Sheet –Power On Stalls & Power Off Stalls(pg 50, memorize these steps)
PHAK Ch. 5; pages 1-13, 22-26, 28-32

What is an aerodynamic stall?

_____

_____

What is the difference between a power off and power on stall?

_____
_____
_____

Rote Knowledge: What is the power-off stall recovery procedure?

_____
_____

Lesson 8, Flight 5: Four Fundamentals and Slow Flight

Date Completed: Instructor Signature:

Lesson Objective: To learn how to perform Slow flight. Since this is the first flight regarding slow flight, it is best to focus on performing the steps of the maneuver while trying to maintain altitude. If done properly, slow flight can be performed without losing any altitude.

Homework:

CTA Stage 1, Phase 3, Section 3
Memory Items Sheet –Slow Flight (pg. 50, memorize)

Why is it important to learn about slow flight?_____
_____
_____

Lesson 9, Ground 5: Airport Operations & Radio Calls

Date Completed: Instructor Signature:

Lesson Objective: To familiarize yourself with the operations at KSUS.

Homework:

PHAK Ch. 14 Airport Ops
AIM Ch. 2, 3 signs & lights
AIM Ch. 4, Section 2, Comms
FAR 91.113 Right of Way, FAR 91.213 ATC Compliance, FAR 91.209 Aircraft Lights
CTA Stage 1, Phase 2, Section 3 & 4

Materials:

Taxiway Light/Marking Handout
Airport Diagram
Traffic Pattern Handout
Phonetic Alphabet Handout

What is the difference between Spirit Tower and Spirit Ground?_____
_____
_____

Lesson 10, Flight 6: Maneuvering in Slow Flight & Stalls

Date Completed: Instructor Signature:

Lesson Objective: To review the slow flight procedure and to learn the power on and power off stall procedures. Emphasis should be on the recovery procedures. Landing shall be performed by the student with assistance from the instructor.

Homework:

CTA Stage 1, Phase 3, Section 1

Memory Items Sheet –Slow Flight, Power On Stall, Power Off Stall (pg. 50, memorize)

What does a power off stall teach us?_____

What does a power on stall teach us?_____

On a separate sheet of paper, write down the procedures for both stalls and the recovery procedure.

Lesson 11, Flight 7: Slow Flight, Stalls, & Steep Turns

Date Completed: Instructor Signature:

Lesson Objective: To review slow flight, power on and power off stalls, and the recovery procedure. Special emphasis should be on the slow flight/power off stall set up and recovery. Landing shall be performed by the student with the instructor's assistance.

Homework:

Maneuver Sheet

Memory Sheet –Steep Turns, Slow Flight, Stalls (pg. 50, memorize)

Lesson 12, Flight 8: Maneuver Review

Date Completed: Instructor Signature:

Lesson Objective: The student shall perform all maneuvers (Steep turns, power on & off stalls, slow flight, associated recoveries), in addition to the takeoff, checklist usage, safety briefing, and pre-maneuver flow. The instructor will act as an evaluator rather than teach, so that the student will have a clear picture of their progress through the course. The only thing the instructor will assist is the landing.

Homework:

Memory Items Sheet –Slow Flight, Stalls, Steep Turns (pg. 50, memorize)

Check In

Ask your instructor the following questions and record it below:

What maneuvers do you feel I am proficient in for my level of training?

What maneuvers do you feel need more work?

What do I need to work on to move forward through the course?

Notes:

Lesson 13, Ground 6: Ground Reference Maneuvers

Date Completed: Instructor Signature:

Lesson Objective: To understand the relationship between the wind and aircraft ground track. Understanding how much bank to use depending on your relationship with the wind is key to performing this maneuver. Make sure to really understand the traffic pattern legs, altitudes, and airspeeds.

Homework:

Memory Items –Ground Reference Maneuvers (pg. 50, memorize)
AFH Ch. 7, Ground Reference maneuvers; Pages 1-10
FAR 91.119 Minimum Safe Altitudes

What is the minimum safe altitude over a congested area? _____

On a separate piece of paper, draw out the traffic pattern for KSUS. Including the altitudes, airspeeds, flaps, and any other applicable information.

Lesson 14, Flight 9: Ground Reference Maneuvers

Date Completed: Instructor Signature:

Lesson Objective: To understand the relationship between the wind and aircraft ground track. Understanding how much bank to use depending on your relationship with the wind is key to performing this maneuver. The student shall be able to identify the legs of the pattern while flying it.

Homework:

On a separate piece of paper, draw out the traffic pattern, legs, altitudes, speeds and flaps. (Repetition is key to memorization).

Memory Items –Ground Reference Maneuvers (pg. 50, memorize)

Lesson 15, Flight 10: Traffic Pattern & Landing

Date Completed: Instructor Signature:

Lesson Objective: To fly the traffic pattern to the standards that were taught. The key is to remain near traffic pattern altitude, associated airspeeds and flap configurations. Landing will also be taught with the instructor's assistance.

Homework:

Draw the Traffic Pattern from Memory
CTA Stage 1, Phase 4, Section 2

Lesson 16, Ground 7: Emergencies

Date Completed: Instructor Signature:

Lesson Objective: To develop good aeronautical decision making using the procedures taught for emergencies. Special emphasis shall go to the Power Loss Procedures (ABC's).

Homework:

AFH Ch. 18, pg. 1-10

Memory Items –Emergencies (pg. 50, memorize)
CTA Stage 1, Phase 3, Section 4

Out of the three skills –Navigate, Communicate, and Aviate- which is the primary skill the pilot should always be performing?
Why?_____
_____
_____
_____
_____

Lesson 17, Flight 11: Emergencies

Date Completed: Instructor Signature:

Lesson Objective: To review the emergency procedures discussed in the previous ground lesson. Emphasis will be placed on the Simulated Engine Out (ABC's) and will be repeated several times so the student may memorize the flow and steps. The student shall perform the entry into the traffic pattern when completing the lesson. If there is time towards the end of the lesson, a few practice landings should be made.

Homework:
CTA Stage 1, Phase 4, Section 4
Memory Items Sheet –Emergencies (pg. 50, memorize)

When would a pilot not use a checklist in an emergency?_____
_____
_____

Lesson 18, Flight 12: Emergencies 2

Date Completed: Instructor Signature:

Lesson Objective: To review the emergency procedures from the previous lesson. The instructor shall demonstrate simulated power off landing. The instructor shall also go over troubleshooting scenarios for other emergency topics. The student shall perform the entry into the traffic pattern when completing the lesson. If there is time towards the end of the lesson, a few practice landings should be made.

What is an example of an emergency that is not covered in the checklist?_____
_____
_____

Lesson 19, Flight 13: Traffic Pattern & Landing

Date Completed: Instructor Signature:

Lesson Objective: To review the traffic pattern and landings. About 10-12 landings shall be made unless the student demonstrates signs of a learning plateau.

Homework:

CTA Stage 1, Phase 1, 4

Why is it important that we land with the main landing gear touching down first?_____

Lesson 20, Ground 9: Aircraft Systems

Date Completed: Instructor Signature:

Lesson Objective: To review aircraft limitations & emergency procedures. The student shall then learn the steps of the four-stroke engine, describe the engine of the 172, the electrical system, fuel system, braking system, and cabin heat. The goal is to have these systems memorized so you could describe them to a 6-year-old, this knowledge is needed for the pre-solo written test.

Homework:

PHAK Ch. 8, systems overview

POH Ch. 7, systems

Draw out the four strokes of a piston engine:

How is the engine cooled? How is it different from your car engine?_____

What does "Direct Drive" mean?_____

How do the spark plugs get their electricity?_____

Where does the engine get its power during the start-up? How about after start-up?_____

_____
_____
___

How many fuel pumps do we have in the Cessna 172 S? _____

How is the electrical system protected? _____

What are the dangers of the Cabin Heat system?_____
_____
_____

Lesson 21, Flight 14: Traffic Pattern & Landings

Date Completed: Instructor Signature:

Lesson Objective: To master the traffic pattern by knowing every leg, altitude, speed and flap setting. Begin to master landings with minimal assistance from the instructor.

Homework:

Draw Traffic Pattern from Memory

How do we decide which runway to use? (26R/8L)_____
_____
_____

Lesson 22, Ground 10: Airspace and Aeronautical Charts 1

Date Completed: Instructor Signature:

Lesson Objective: To become familiar with sectional charts and the airspace system. Emphasis should be placed on the airport of intended solo; that information will be tested on in the pre-solo written.

Homework:

PHAK Ch. 15 Airspace
Sectional Chart
CTA Stage 1, Phase 4, Section 1

What class of airspace do we operate out of? How can you tell?_____
_____
____

What equipment do we need in our airspace?_____
_____
_____

What is the length of our longest runway? _____

What frequencies do we need to know for a trip in the pattern starting on the ramp?_____

Lesson 23, Flight 15: Traffic Pattern & Landings

Date Completed: Instructor Signature:

Lesson Objective: To fly the traffic pattern with minimal instructor input, with assistance on the landing.

Homework:

CTA Stage 1, Phase 5, Section 1

What inputs must be made to land with a right crosswind?_____

Lesson 24, Ground 11: Weight & Balance, Performance

Date Completed: Instructor Signature:

Lesson Objective: To understand why we must perform a Weight and Balance and how to do so. The student should be able to complete and W&B/Performance sheet on their own by the end of the lesson.

Materials:

Weight and Balance Sheet (located in CFI room under the FRAT sheets)

Homework:

CTA Stage 1, Phase 3, 2
POH Ch. 6, Weight & Balance
POH Ch. 5, Performance
PHAK Ch. 10

What is our maximum takeoff weight? What happens if we exceed it?_____

Why do we have to ensure the aircraft is balanced?_____

What C.G. position is better for cruise performance? Why?_____

_____
_____

Lesson 25, Flight 16: Traffic Pattern & Landings

Date Completed: Instructor Signature:

Lesson Objective: To fly the pattern with little to no instructor assistance, including the landing.

What is ground effect?_____
_____
_____

What causes us to "float" in ground effect? What do we do when that happens?_____
_____
_____

Lesson 26, Ground 19: Aircraft Inspections, Inop Equipment & Documents

Date Completed: Instructor Signature:

Lesson Objective: To recite the TOMATO FLAMES acronym and be able to explain the process when a component is inoperative. This subject is heavily emphasized in our curriculum, extra time should be taken to fully understand this concepts.

Materials:

POH Ch. 6 – Comprehensive Equipment List

Homework:

PHAK Ch. 9, Airworthiness

What makes an aircraft airworthy?_____
_____
_____

How do we maintain an airworthiness certification?_____
_____
_____
_____

What would we do if we found that the attitude indicator wasn't working?_____
_____
_____
_____
_____

What would we do if there was a small bald spot on the tire?_____

Lesson 27, Flight 17: Traffic Pattern & Landings

Date Completed: Instructor Signature:

Lesson Objective: To review pattern work and landings with little to no assistance from the instructor. The student should be making radio calls at this point. The instructor shall give you the pre-solo written test. It is an open book test that you may take home and fill out.

Homework:

V Speeds from memory

What is the Go-Around procedure?_____

When should we perform a Go-Around?_____

Lesson 28, Flight 18: Review

Date Completed: Instructor Signature:

Lesson Objective: To review all maneuvers including slow flight, power on and power off stall with recoveries, steep turns, emergency procedures, forward slip, traffic pattern and landing. Emphasis is on the Go-Around procedure, stall recovery/indication of a stall, traffic pattern and landings. The goal is to perform all these maneuvers with no assistance from the instructor.

What does a forward slip allow us to do?_____

Aerodynamically, how does a forward slip work?_____

Lesson 29, Ground 12: Oral Review

Date Completed: Instructor Signature:

Lesson Objective: To be able to review: V-speeds & Limitations, local airspace, local airport operations, METARs, Emergencies, pattern operations, stall & spin awareness, pre-maneuver flow, power on and power off stalls with recoveries, steep turns, and landings including crosswind. The student should be able to discuss all topics from memory and demonstrate a good understanding of procedure and theory. Review your pre-solo written test results with your instructor.

Why is it important for you (the student) to understand the above concepts?_____
_____
_____

Check In

Ask your instructor the following questions and record it below:

What about my current performance do you feel is ready to solo?

What about my current performance makes you hesitate to send me for solo?

What steps do we need to take to get me prepared for solo?

Lesson 30, Ground 13: Pre-solo Prog Check Oral

Date Completed: Instructor Signature:

Lesson Objective: To discuss with a stagecheck instructor the topics covered in the previous lesson.

Lesson 31, Flight 19: Pre-solo Prog Check

Date Completed: Instructor Signature:

Lesson Objective: To demonstrate good knowledge of procedure, situational awareness, and aeronautical decision making in the local area and in the pattern. The student should be able to perform everything on their own with no assistance from the stagecheck instructor.

Check In

Ask your stagecheck instructor the following questions and record it below:

If I were your student, would you send me solo? Why or why not?

What would you recommend I work on overaoral review, is there any topics I need to improve upon?

Pertaining to my oral, are there any subject areas that need improvement?

Lesson 32, Flight 20: Pattern & First Solo

Date Completed: Instructor Signature:

Lesson Objective: To solo in the aircraft. The instructor may determine you require extra flights at this point if the consistency and technique need improvement before solo. This is up to the instructor's discretion. Homework must be completed before flight.

Homework:

Pre-solo written test (ask your instructor for it)

Lesson 33, Ground 14: Navlogs, Chart Supplements, & Aeronautical Charts 2

Date Completed: Instructor Signature:

Lesson Objective: To review sectional charts and airspace. In addition, you will learn the basics of Navlogs and how to plan a cross country.

Homework:

FAR 91.159
CTA Stage 12, Phase 2, Section 1
Practice written test results must be shown to CFI

Materials:

Sectional Chart
Plotter & E6B
Navlog
Pen or Pencil

What altitude should we fly if we're going from KSUS to KDEN?_____

What is our minimum fuel requirement?

_____

What is the purpose of the Navlog?_____

Lesson 34, Ground 15: Short & Soft Field Takeoffs & Landings

Date Completed: Instructor Signature:

Lesson Objective: To discuss the purpose of short and soft field landings and takeoffs. The student shall review the short and soft procedures found in this notebook and commit them to memory.

Homework:

POH. Ch 5, Performance
Elite Maneuver Notebook

CTA Stage 2, Phase 1, Section 1
Have a new Navlog filled out to review

What is the soft field takeoff procedure?_____

Lesson 35, Flight 21: Short & Soft Field Operations

Date Completed: Instructor Signature:

Lesson Objective: To perform the maneuvers discussed in the previous lesson.

Homework:
CTA Stage 2, Phase 2, Section 2
Practice Written Test is due –See Cessna Training Application for practice written.

Why do we hold the nose off in a soft field landing?_____

Lesson 36, Flight 22: Cross Country Operations 1

Date Completed: Instructor Signature:

Lesson Objective: The student shall navigate to and from an airport 50nm away or greater. The student shall use pilotage to identify the predetermined checkpoints. The student shall only use the paper sectional chart and paper navlog for navigation. No GPS usage.

Homework:
CTA Stage 2, Phase 7, 1
Navlog –prepare prior to your lesson
Weight and Balance –prepare prior to your lesson

What is the difference between pilotage and dead reckoning?_____

Lesson 37, Ground 15: Weather Reporting Systems & Briefs

Date Completed: Instructor Signature:

Lesson Objective: To learn about the various weather reporting systems, how to read them, why they're important, and how to use them for planning your cross countries.

Homework:
PHAK Ch. 13 Aviation Weather Services

CTA Stage 2, Phase 1, 2
Practice Written test results must be shown to CFI

What is the difference between a METAR and a TAF?_____

Where can you get an appropriate weather briefing?_____

What's the difference between an AIRMET, SIGMET, and Convective SIGMET?_____

Lesson 38, Flight 23: Cross Country Operations 2

Date Completed: Instructor Signature:

Lesson Objective: The student shall perform another cross country of 50nm or greater. This time using both pilotage and dead reckoning. The student shall only use the paper sectional and paper navlog for navigation. No GPS usage.

Homework:

CTA Stage 2, Phase 3, 2
Navlog –prepare prior to your lesson
Weight and Balance –prepare prior to your lesson

Lesson 39, Ground 16: Weather Theory

Date Completed: Instructor Signature:

Lesson Objective: The student should have a thorough understanding of weather theory particularly Stable/unstable air masses and how they relate to thunderstorms, fronts, and pressure systems. Icing, the causes of icing, and how to recover should also be discussed.

Homework:

PHAK Ch. 12, Weather Theory
CTA Stage 2, Phase 4, Section 1
Practice Written test results must be shown to CFI –70% or above

How would you describe stable air as opposed to unstable air?_____

_____
_____

What are the ingredients of a thunderstorm?_____
_____
_____

Lesson 40, Flight 24: Cross Country Operations 3

Date Completed: Instructor Signature: _____

Lesson Objective: The student shall use pilotage and dead reckoning to a destination 50nm or greater. The student shall also use only paper charts and paper navlogs. No GPS. The goal of this lesson is for the student to be able to navigate on their own with no help from the instructor.

Homework:

Navlog –prepare prior to your lesson

Weight and Balance –prepare prior to your lesson

What are the weather minimums for class E airspace?_____
_____
_____

Lesson 41, Ground 17: Basic Attitude Instrument Flight & Unusual Attitudes

Date Completed: Instructor Signature:

Lesson Objective: To understand the basics of an instrument scan and how it applies to Instrument Meterological Conditions (IMC). The student should also understand what an "unusual attitude" is and how to correct for them.

Homework:

Practice Written test results must be shown to CFI –75% or above

What are the pitot-static instruments?_____

What powers the vacuum pump?_____

What instruments utilize the vacuum pump and how do they do so?_____

Lesson 42, Ground 18: Night Operations & Aeromedical Factors

Date Completed: Instructor Signature:

Lesson Objective: To understand aeromedical factors, symptoms, and how to correct those symptoms. Emphasis shall be placed on the topics mentions in the Airmen Certification Standards task that covers this topic.

Resources:

AFH, Ch. 11
PHAK Ch. 17, Aeromedical Factors
CTA Stage 2, Phase 3, Section 1
Practice Written test results must be shown to CFI –80% or above

What type of hypoxia is carbon monoxide poisoning? _____

Why don't we see color at night?_____

What hazardous attitude do you expect is the most dangerous? Why?_____

_____

What is loggable nighttime?

_____

Lesson 43, Flight 25: Night Takeoffs & Landings

Homework:

- Practice written test results must be shown to CFI –85% or above

Date Completed: Instructor Signature:

Lesson Objective: To apply the night operations techniques discussed in the previous lesson. This lesson will focus on takeoffs and landings at the local airport during night conditions.

Why is it important that we use our peripheral vision while landing at night?

_____
_____
_____

Lesson 44, Flight 26: Night Cross Country

Date Completed: Instructor Signature:

Lesson Objective: To fly a cross country at night, and further apply the principles discussed in the last two lessons. The student shall use pilotage and dead reckoning to navigate with no GPS usage and paper charts.

Homework:

CTA Stage 2, Phase 8, 1
Practice written test results must be shown to CFI -90% or above

Why is it important for the student to use paper charts and no GPS to learn to navigate?

_____
_____
_____
_____

Lesson 45, Sim 1: Lost Procedures, VOR Navigation & GPS Operation

Date Completed: Instructor Signature:

Lesson Objective: To memorize and understand the Lost Procedures and basic VOR navigation. The student will also learn how to cross-reference VOR radials to find their position. GPS Direct-To function will also be discussed in this lesson.

Homework:

Practice Written test results must be shown to CFI –above 90%

What is a VOR? How does it work?_____
_____
_____
_____

What's the difference between the TO and FROM feature of the OBS?_____

How do we get TO a VOR?_____

Lesson 46, Flight 27: Lost Procedures, Diversions

Date Completed: Instructor Signature:

Lesson Objective: To reiterate the procedures covered in the previous sim lesson. The student should be able to perform VOR, GPS, and lost procedures while flying the airplane. The goal is that the instructor must step in as little as possible, if at all.

Homework:

Practice Written test results must be shown to CFI –above 90%

Where can we find information about the operation of the autopilot?

_____

Lesson 47, Flight 28: Cross Country Operations, Lost Procedures & Diversions

Date Completed: Instructor Signature:

Lesson Objective: The goal of this lesson is to perform a diversion. The instructor may choose when and what scenario to divert. The student must navigate using pilotage and dead reckoning; no GPS shall be used for this diversion.

Homework:

Written must be scheduled within a week of this lesson

Why is it an important skill to know how to divert without using the GPS?_____

Check In

Ask your instructor the following questions and record it below:

What about my current performance do you feel is ready to solo xc?

What about my current performance makes you hesitate to send me for solo xc?

What steps do we need to take to get me prepared for solo xc?

In what subject areas do I need to improve my knowledge and understanding?

Written is due –Do not continue past this point

Lesson 48, Ground 18: Pre-solo XC Oral Prog Check

Date Completed: Instructor Signature:

Lesson Objective: To discuss all topics covered up until this point. Special emphasis shall be placed on a cross-country flight plan. The scenario for this cross country will be given to you by your instructor. The student must plan this on a paper navlog and use paper charts. Lost procedures, diversions, and VOR operation will also be discussed. It is up to the student to be familiar with all the details concerning the flight, this includes Weight and balance, aircraft documents and inspections, current and forecast weather, and information concerning the airport of intended landing.

Note: This prog check oral closely mimics how your checkride oral will be performed. Pay attention to what subject areas you have trouble addressing so that you may improve them with your instructor.

Lesson 49, Flight 29: Pre-solo XC Prog Check

Date Completed: Instructor Signature:

Lesson Objective: To demonstrate the cross country previously discussed. A diversion, BAI, and lost procedures may be expected from the check instructor.

Check In

Ask your stagecheck instructor the following questions and record it below:

What about my current performance do you feel is ready to solo xc?

What about my current performance makes you hesitate to send me for solo xc?

What do my instructor and I need to work on before my solo xc?

Lesson 50, Flight 30: First Solo XC

Date Completed: Instructor Signature:

Lesson Objective: To perform a solo xc to an airport 50 nm or greater. **Minimum 1.5 hours of flight.**

Homework:

Navlog
Weight and Balance

Minimum Flight Time: 1.5 hours

Lesson 51, Flight 31: Second Solo XC

Date Completed: Instructor Signature:

Lesson Objective: To perform a solo xc to an airport 50 nm or greater. **Minimum 1.5 hours of flight.**

Homework:

Navlog
Weight and Balance

Minimum Flight Time: 1.5 hours

Lesson 52, Flight 32: Long Solo XC

Date Completed: Instructor Signature:

Lesson Objective: To perform a solo xc to an airport 50 nm or greater. **Minimum 1.5 hours of flight**.

Homework:

CTA Stage 2, Phase 9, 1
Navlog
Weight and Balance

What is the minimum solo xc time required for a private pilot's license?_____

How much time do you currently have? Do you need more solo xc time?_____

Minimum Flight Time: 1.5 hours

Lesson 53, Flight 33: Dual Maneuvers Evaluation

Date Completed: Instructor Signature:

Lesson Objective: To review maneuvers after a period of not performing them. The instructor will be evaluating rather than instructing. The goal is to perform all maneuvers to Private Pilot Airman Certification Standards (ACS). The instructor will provide feedback on maneuvers so that the student may solo in the future and be able to improve.

Homework:

CTA Stage 3, Phase 1, Section 1

Where can you find the Private Pilot ACS?_____

What is the purpose of the ACS?_____

Check In

Ask your instructor the following questions and record it below:

What about my current performance do you feel is ready for the End of Course Stagecheck?

What about my current performance makes you hesitate to send me for the End of Course Stagecheck?

Lesson 54, Ground 20: Private Pilot Privileges and Limitations

Date Completed: Instructor Signature:

Lesson Objective: For the student to understand the FAR/AIM and what regulations apply to them as private pilots.

Resources:

FAR 61.83, 61.89, 61.23, 91.155

Homework:

CTA Stage 3, Phase 1, Section 2

What are the privileges and limitations of a private pilot?_____

Lesson 55, Flight 34: Solo to the Practice Area 1

Date Completed: Instructor Signature:

Lesson Objective: To review maneuvers determined by your last dual flight.

What maneuver will you focus on? _____

What will you do to improve
it?_____
_____
__

Lesson 56, Flight 35: Solo to the Practice Area 2

Date Completed: Instructor Signature:

Lesson Objective: To review maneuvers.

What are the completion standards as determined by the ACS for steep
turns?_____
____

If you lose altitude during a steep turn, what must you do to
compensate?_____
_____

Lesson 57, Flight 36: Solo to the Practice Area 3

Date Completed: Instructor Signature:

Lesson Objective: To review maneuvers and perform them within ACS.

Why does bank change during a turn around a
point?_____
_____
_____

Lesson 58, Ground 21: Mock Oral

Date Completed: Instructor Signature:

Lesson Objective: To stage an oral like the End of Course oral to discover deficiencies in ground knowledge. This is not to be done with an End of Course instructor. This covers all topics in the ACS.

Homework:

CTA Stage 3, Phase 10, 1&2

End of Course Written Test will be assigned. Disclaimer: The End of Course written test is a take-home test designed to address the memorized knowledge required for the checkride. We want to identify gaps in knowledge prior to signing up for the End of Course Ground test. It is to your advantage to take the test by using the guidelines provided on the test. Any evidence of straying from said guidelines may prompt your instructor to administer the test during a ground lesson. Here are the guidelines for reference:

EOC Written Test Guidelines

- Start by answering as many questions as you can from memory.

- Go over the questions you can't answer from memory and use appropriate aviation sources to answer. Be sure to mark any questions you have to look up with a different colored pen.
- It is to your advantage to follow these directions, as it will aid in your training and checkride process. However, if there is evidence that you did not follow these steps, your CFI does have the right to make you take the whole test in front of them from memory.

Check In

Ask your instructor the following questions and record it below:

What about my current knowledge base do you feel is ready for the End of Course Stagecheck?

What about my current knowledge base makes you hesitate to send me for the End of Course Stagecheck?

What topics do I need to improve for this upcoming test?

What about those topics need review? Is it a lack of knowledge, understanding, application or correlation?

Lesson 59, Flight 37: Mock Checkride Flight

Date Completed: Instructor Signature:

Lesson Objective: To cover all topics on the ACS and perform them at the standard or better.

Check In

Ask your instructor the following questions and record it below:

What maneuvers or areas need improvement?

Based on my performance today, what would make you hesitate to send me for an End of Course Stagecheck?

What about those performance items need review? Is it a lack of knowledge, understanding, application or correlation?

What are my next steps to ensure I pass the End of Course Stagecheck?

Note: At this point in your training your instructor may deem it necessary to repeat some lessons until a level of proficiency is reached.

Lesson 56, Ground 22: Logbook Review

Date Completed: Instructor Signature:

Lesson Objective: To review the aircraft maintenance logs and your own personal logbook for any deficiencies. For your logbook, you must ensure you meet all the minimum requirements to apply for the license. If these requirements aren't met, please talk to your instructor about what you need to complete them. Your written test deficiencies will also be discussed in this lesson, so please **bring your written test results** with you.

Documents to complete before End of Course:

- Checkride Checklist
- EOC PPL Written Test
- Pilot Logbook Tabbed
- Aircraft Maintenance Records Tabbed
- Written Test Deficiencies Endorsement

Note: In addition to the minimum requirements, we are ensuring your safety by performing all line items in the ACS to standard. If the instructor deems it necessary, you may need to repeat some lessons to achieve that proficiency.

Check In

Ask your instructor the following questions and record it below:

Based on the minimum time required, am I deficient in any areas?

If so what are they and what do we need to do to correct for them?

End of Course Ground

Date Completed: Instructor Signature:

Lesson Objective: To take the End of Course ground with one of our designated End of Course Instructors. This is a Pass/Fail event.

List of required documents to bring on your End of Course:

- Student Logbook (tabbed out)
- Photo ID
- Medical
- Student Pilot Certificate
- FAA Written Test Results
- EOC Written Test
- Checkride Checklist Completed
- Aircraft Maintenance Records (tabbed out)

Check In

Ask your End of Course instructor the following questions and record it:

What about my current knowledge base do you feel is ready for the FAA Checkride?

What about my current knowledge base makes you hesitate to send me for the FAA Checkride?

What topics do I need to improve for this upcoming test?

What about those topics need review? Is it a lack of knowledge, understanding, application or correlation?

End of Course Flight

Date Completed: Instructor Signature:

Lesson Objective: To take the End of Course flight with one of our designated End of Course Instructors. This is a Pass/Fail event.

Note: If the EOC instructor determines the flight or oral portion of this test to be a “Fail” event, the student must seek remedial instruction from their assigned instructor. The EOC will then have to be retaken until all line items are marked satisfactory and within ACS standards.

Check In

Ask your End of Course instructor the following questions and record it:

What about my performance do you feel is ready for the FAA Checkride?

What about my current knowledge base makes you hesitate to send me for the FAA Checkride?

What topics do I need to improve for this upcoming test?

What about my performance needs review? Is it a lack of knowledge, understanding, application or correlation? For which performance areas?

Private Pilot Maneuvers

Cessna 172-S

TAKEOFF:

Normal Takeoff

- Complete the Before Takeoff Checklist and review the Takeoff Checklist
- Turn applicable lights on, set transponder to ALT mode, check wind indicator, and ensure clearance (if applicable) prior to taxiing onto the active runway.
- When aligned with runway centerline, verify heading indicator, note time, smoothly add full power
- Check tachometer to verify full power is being developed by the engine, airspeed indicator is alive, and engine instruments are in the green
- Rotate at 55 KIAS (Vr)
- Pitch to establish 74 KIAS (Vy) after leaving the runway surface

Cross Wind Takeoff

- Complete the Before Takeoff Checklist and review the Takeoff Checklist
- Turn applicable lights on, set transponder to ALT mode, and check wind indicator, and ensure clearance (if applicable) prior to taxiing onto the active runway.
- When aligned with runway centerline, turn ailerons fully into the wind, verify heading indicator, note time, and smoothly add full power
- Check tachometer to verify full power is being developed by the engine, airspeed indicator is alive, and engine instruments are in the green
- During takeoff roll, reduce aileron deflection as needed to maintain runway centerline
- Rotate at 55 KIAS (Vr)
- Pitch to establish 74 KIAS (Vy) after leaving runway surface
- Maintain the extended runway centerline through an appropriate crab angle, or sideslip

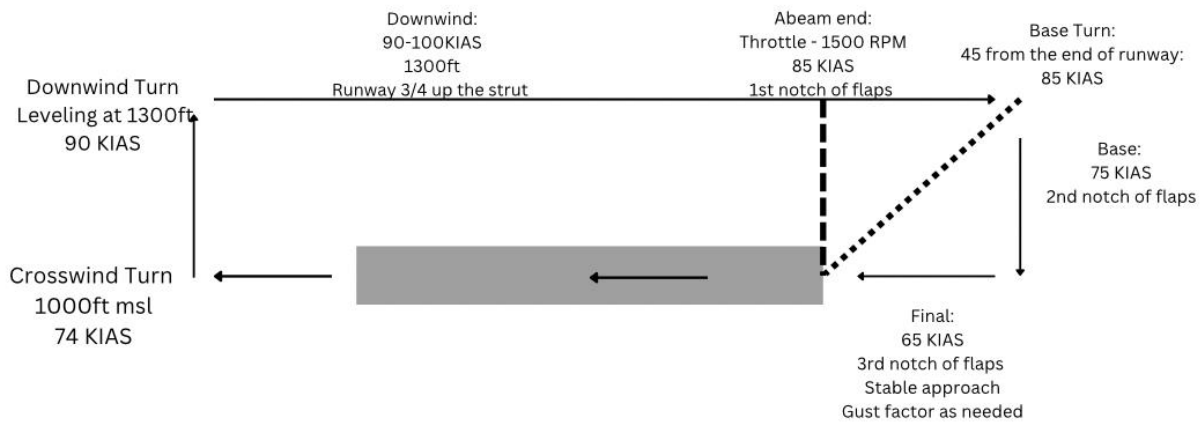
Short Field Takeoff

- Complete the Before Takeoff Checklist and review the Short Field Takeoff Checklist
- Deploy 10° of flaps, turn applicable lights on, set transponder to ALT mode, and check wind indicator, and ensure clearance (if applicable) prior to taxiing onto the active runway.
- Taxi to the runway end using all available pavement
- Align yourself with the runway centerline, moving forward just enough to allow the nose wheel to straighten and apply full brakes
- Verify heading indicator, note time, and smoothly apply full power
- Check tachometer to verify full power is being developed by the engine, and engine instruments are in the green
- Release the brakes
- Verify airspeed indicator is alive
- Rotate at 50 KIAS and then pitch to establish 56 KIAS (Vx)
- Once at 50' AGL, or when advised, and when all obstacles are cleared, pitch to establish a climb at 74 KIAS (Vy)
- Above 60 KIAS retract the flaps

Soft Field Takeoff

- Complete the Before Takeoff Checklist and review applicable items from the Takeoff Checklist
- Deploy 10° of flaps, Turn applicable lights on, set transponder to ALT mode, and check wind indicator, and ensure clearance (if applicable) prior to taxiing onto the active runway.
- Taxi onto the runway with full aft elevator pressure and align on runway centerline without coming to a full stop
- Smoothly apply full power, once the nose wheel is off the runway surface, begin to reduce backpressure to maintain forward visibility while keeping the nose wheel off the runway
- Check tachometer to verify full power is being developed by the engine, airspeed indicator is alive, and engine instruments are in the green
- Once airborne, level off in ground effect until the aircraft reaches 74 KIAS (Vy) (Or Vx if appropriate), then climb out maintaining that speed (Vy or Vx as appropriate)
- Above 100' AGL (or all obstacles clear) and above 60 KIAS and positive rate of climb, retract the flaps

LANDINGS:



Normal Landing

- Complete the Before Landing and Landing Checklists
- Descend to pattern altitude ~3 miles prior to entering traffic pattern, or as appropriate
- At midfield downwind perform the CGUMPS procedure:
  - C – Clearance: permission to land/option from ATC if applicable
  - G – Gas: Fuel selector valve on both
  - U – Undercarriage: Verify gear down (or fixed gear)
  - M – Mixture: Full rich
  - P – Power: reduce to establish 100 KIAS
  - S – Seatbelts: latched and locked in place for all occupants
- Abeam the touch down point, reduce power to 1500 RPM, or as appropriate
- Deploy 10° of flaps when at or below 110 (Vfe)
- Pitch to establish descent at 80 KIAS and trim as necessary
- On base, deploy flaps to 20° (as appropriate) **ensuring under 85 KIAS**, pitch to establish 75 KIAS and adjust trim as necessary
- On final, re-verify wind conditions, maintain runway centerline, deploy flaps to 30° (as appropriate), pitch to establish 65 KIAS (+ ½ the gust factor) and trim as necessary
- Reduce power as needed, and reduce to idle when runway is assured

Cross Wind Landing

- Complete the Before Landing and Landing Checklists
- Descend to pattern altitude ~3 miles prior to entering traffic pattern, or as appropriate
- At midfield downwind perform the CGUMPS procedure:
  - C – Clearance: permission to land/option from ATC if applicable
  - G – Gas: Fuel selector valve on both
  - U – Undercarriage: Verify gear down (or fixed gear)
  - M – Mixture: Full rich
  - P – Power: reduce to establish 100 KIAS
  - S – Seatbelts: latched and locked in place for all occupants
- Abeam the touch down point, reduce power to 1500 RPM, or as appropriate
- Deploy 10° of flaps when at or below 110 (Vfe)
- Pitch to establish descent at 80 KIAS and trim as necessary
- On base, deploy flaps to 20° (as appropriate) **ensuring under 85 KIAS**, pitch to establish 75 KIAS and adjust trim as necessary
- On final, re-verify wind conditions, establish sideslip by applying ailerons into the wind as needed and opposite rudder as needed to maintain runway centerline
- Pitch to establish 65 KIAS (+ ½ the gust factor) and trim as necessary
- Reduce the power as needed, and reduce the power to idle when runway is assured, while maintaining the sideslip and centerline
- Touch down on the upwind main gear first, followed by the opposite main gear, and then the nose gear
- After touchdown, gradually turn ailerons fully into the wind

Short Field Landing

- Complete the Before Landing and Short Field Landing Checklists
- Descend to pattern altitude ~3 miles prior to entering traffic pattern, or as appropriate
- At midfield downwind perform the CGUMPS procedure:
  - C – Clearance: permission to land/option from ATC if applicable
  - G – Gas: Fuel selector valve on both
  - U – Undercarriage: Verify gear down (or fixed gear)
  - M – Mixture: Full rich
  - P – Power: reduce to establish 100 KIAS
  - S – Seatbelts: latched and locked in place for all occupants
- Select and announce the desired touch down point
- When abeam selected touch down point reduce power to 1500 RPM, as appropriate
- Deploy 10° of flaps when at or below 110 (Vfe)
- Pitch to establish descent at 80 KIAS and trim as necessary
- On base, deploy flaps to 20° (as appropriate) **ensuring under 85 KIAS**, pitch to establish 75 KIAS and adjust trim as necessary
- On final, re-verify the wind conditions, maintain runway centerline, deploy flaps to 30° (as appropriate), pitch to establish 61 KIAS and trim as necessary
- Adjust power as needed and reduce power to idle as appropriate to land on or shortly after the identified touch down point

- Upon touch down, apply smooth and equal braking without locking the brakes, add backpressure, and retract the flaps for maximum brake effectiveness

Soft Field Landing

- Complete the Before Landing and Short Field Landing Checklists
- Descend to pattern altitude ~3 miles prior to entering traffic pattern, or as appropriate
- At midfield downwind perform the CGUMPS procedure:
  - C – Clearance: permission to land/option from ATC if applicable
  - G – Gas: Fuel selector valve on both
  - U – Undercarriage: Verify gear down (or fixed gear)
  - M – Mixture: Full rich
  - P – Power: reduce to establish 100 KIAS
  - S – Seatbelts: latched and locked in place for all occupants
- Abeam the touch down point, reduce power to 1500-1700 RPM as appropriate
- Deploy 10° of flaps when at or below 110 (Vfe)
- Pitch to establish descent at 80 KIAS and trim as necessary
- On base, deploy flaps to 20° (as appropriate) ensuring under 85 KIAS and pitch to establish 75 KIAS and adjust trim as necessary
- On final, re-verify the wind conditions, maintain runway centerline, deploy flaps to 30° (as appropriate), pitch to establish 65 KIAS and trim as necessary
- Reduce the power as needed and when the runway is assured, reduce the power to just above idle, flare, and land softly;
  - **OR** reduce the power to idle as appropriate and add power as needed during the flare to control the rate of descent and land softly
- Touch down gently on the main gear while maintaining backpressure to allow the nose gear to remain off of the runway until it drops naturally, using MINIMAL braking

Power Off 180 Landing (AKA Short Approach) **<u>(With instructor only)</u>**

- Reduce throttle to idle smoothly when abeam the point of intended landing on downwind or when instructed
- Pitch to establish glide at 68 KIAS (Vg)
- Trim as necessary
- **Turn base at an appropriate distance from the runway based on the strength and direction of the wind**
- Deploy up to 30° of flaps as appropriate only when runway is assured
- On final, re-verify the wind conditions, maintain 68 KIAS (Vg) and trim as necessary
- **<u>NEVER</u>** attempt to stretch the glide or retract the flaps to reach the point of intended landing

Go-Around

- Immediately and simultaneously:
  - Smoothly apply full power
  - Pitch slightly up to achieve and maintain a safe airspeed – above 60 KIAS

    - Retract flaps to 20°
- After gaining a positive rate of climb and maintaining a safe airspeed, retract flaps to 10°
- Once clear of obstacles, at a safe altitude, and above 65 KIAS, retract to 0° of flaps and climb as normal at 74 KIAS (Vy)
- Inform ATC or traffic of go-around
    - RECALL: (Aviate, Navigate, Communicate; in that order)

Forward Slips to Land

- **Use of flaps during slips are prohibited unless with a CFI, or in the event of an emergency**
- **Do not induce a skidding turn:** Use of slips during turns not recommended
- Reduce throttle appropriately for altitude
- Apply opposing aileron and rudder inputs to maintain track toward centerline, and forward elevator input to maintain pitch down attitude
- When on appropriate glidepath and landing point is assured, remove or reduce the opposing control inputs to continue a normal approach to landing or sideslip as appropriate

MANEUVERS:

Steep Turns

- Maintain an altitude above 1500' AGL
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – adjust as appropriate to establish entry airspeed of 95 KIAS
    - Mixture – full rich if below 3000' MSL
    - Flaps – clean configuration
    - Fuel Selector Valve – both position
- Roll into a 45° bank
- Perform a 360° turn in either direction
- Adjust power as necessary to maintain airspeed
- Adjust backpressure as appropriate to maintain altitude
- Nose up trim as appropriate
- Roll out on entry heading, adjusting power, backpressure, and/or trim as necessary to maintain altitude and airspeed
- Repeat in the opposite direction of turn
- Maintain coordination throughout the maneuver

Slow Flight

- Maintain an altitude above 1500’ AGL
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – reduce the power to 1500 – 1700 RPM
    - Mixture – full rich if below 3000’ MSL
    - Flaps – Deploy 10° of flaps when at or below 110 KIAS (Vfe)
    - Fuel Selector Valve – both position
- Deploy the remaining flaps in 10° increments once below 85 KIAS.
- Pitch to establish and maintain desired airspeed
- Adjust power as appropriate to maintain altitude
- Trim as necessary
- Maintain coordination throughout the maneuver
- Recovery:
    - Smoothly add full power while maintaining original altitude
    - Allow the airspeed to increase
    - Retract flaps to 20°
    - Above 65 KIAS, retract the remaining flaps in 10° increments
    - Above 100 KIAS, return to a cruise flight configuration per Cruise Checklist

Power Off Stall

- Maneuver is to be completed at an altitude above 1500’ AGL
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – reduce the power to 1500 – 1700 RPM
    - Mixture – full rich
    - Flaps – Deploy 10° of flaps when at or below 110 KIAS (Vfe)
    - Fuel Selector Valve – both position
- Maintain altitude with backpressure and allow the airspeed to fall below 85 KIAS
- Deploy the remaining flaps in 10° increments
- Establish a controlled descent at 65 KIAS for 200’
- Maintain wings level or establish up to a 20° bank as instructed
- Maintain coordination throughout the maneuver
- Reduce the power to idle and add backpressure until the critical angle of attack is reached
- Recognize and announce vocally the horn, buffet, and stall
- Recovery:
    - Decrease the angle of attack and smoothly add full power
    - Maintain coordination throughout

    - Retract the flaps to 20°
    - Establish a climb
    - When above 65 KIAS, retract flaps to 10°
    - When above Vx or Vy as appropriate, retract the last 10° of flaps.
    - Upon leveling off on starting or specified altitude, return to cruise flight per Cruise Checklist

Power On Stall

- Maneuver is to be completed at an altitude above 1500' AGL
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – reduce the power to 1500 – 1700 RPM
    - Mixture – full rich
    - Flaps – clean configuration
    - Fuel Selector Valve – both position
- Maintain altitude with backpressure and allow the airspeed to reach 55 KIAS
- Maintain wings level or establish up to a 20° bank as instructed
- Maintain coordination throughout the maneuver
- At 55 KIAS (Vr) add full power and backpressure until the critical angle of attack is reached.
- Recognize and announce vocally the horn, buffet, and stall
- **Recovery:**
    - Decrease the angle of attack
    - Maintain coordination throughout
    - Establish a climb
    - Upon leveling off on starting or specified altitude, return to cruise flight per Cruise Checklist

GROUND REFERENCE MANEUVERS:

Rectangular Course

- Select and maintain an altitude between 600’ – 1000’ AGL
- Find an area that is clear of congestion, obstruction, and terrain
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – cruise setting
    - Mixture – full rich
    - Flaps – clean configuration
    - Fuel Selector Valve – both position
- Enter the maneuver at a 45° angle to the downwind
- Maintain a constant distance from the course by crabbing into the wind and varying the bank angle during turns based on the ground track
- Maintain coordination throughout the maneuver

Turns Around a Point

- Select and maintain an altitude between 600’ – 1000’ AGL
- Find an area that is clear of congestion obstructions and terrain
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position
    - Throttle – cruise setting
    - Mixture – full rich
    - Flaps – clean configuration
    - Fuel Selector Valve – both position
- Recommended entry from downwind
- Based on the ground speed, maintain a constant distance away from the ground reference point by varying bank angle
- Perform two complete circles around the ground reference point
- Maintain coordination throughout the maneuver

S-Turns

- Maintain an altitude between 600’ and 1000’ AGL
- Find an area that is clear of congestion obstructions and terrain
- Accomplish clearing turns, either one 180° turn or two 90° turns
- Perform the Pre-Maneuver Flow:
    - Ignition switch – both position
    - Master switch – both position

    - Throttle – cruise setting
    - Mixture – full rich
    - Flaps – clean configuration
    - Fuel Selector Valve – both position
- Find a ground reference line (road, railroad, field line, tree line) that runs perpendicular to the wind
- Recommend entry from downwind
- Cross your reference line with wings level
- Roll into a bank not to exceed 45° and vary bank angle based on groundspeed to create a perfect semi-circle
- Roll out wings level when crossing reference line
- Repeat the maneuver in the opposite direction of turn to create a second perfect semi-circle

EMERGENCY PROCEDURE PRACTICE:

Simulated Engine Failure (With instructor only)

- Smoothly reduce power to idle
- Pitch and trim as necessary to establish 68 KIAS (Vg)
- Identify best field of intended landing
- **NOTE:** Vocalize but do not perform actions contrary to safety or that will result in the engine or electrical system shutting off
- Perform Engine Failure During Flight Checklist
- If engine fails to restart, perform the Landing Without Engine Power Checklist
- Transponder – simulate/mention setting to 7700
- Simulate a distress call to ATC on a nearby frequency or on emergency frequency 121.5 (Do not transmit)
- Recovery:
    - Between 600-1000' AGL
    - Smoothly add full power
    - Pitch to establish 74 KIAS (Vy)
    - Retract flaps in 10° increments if they were deployed
    - Trim as necessary

Simulated Emergency Descent (With instructor only)

- Smoothly reduce power to idle
- Pitch to establish airspeed at 100 KIAS (for training purposes)
- Establish up to 45° of bank as instructed while in a descent
- Trim as appropriate
- Recovery:
    - Establish wings level if in a descending turn
    - Smoothly increase backpressure to establish a level flight attitude
    - Smoothly increase power as necessary
    - Fully recover at an altitude above 1000' AGL

Memory Items Sheet

To simplify procedures, we have a Memory Items sheet. The sheet takes every maneuver and condenses it to just a few steps, making it easier to memorize. The real key to memorizing maneuvers, however, is by repetition. Not just in the airplane, but the act of writing and re-writing the maneuvers from the Memory Items sheet. Items highlighted in Bold **must be memorized.**

Pre-Maneuver Flow

C – Clearing Turns
A – Airspeed (Appropriate for Maneuver)
S – Safe place to Land (In event of emergency)
A –Altitude (Appropriate for Maneuver)

Slow flight

- **Pre-Maneuver Flow**
- **Power –1500RPM**
- **1st notch below 110kts**
- **2nd & 3rd notch below 85kts**
- **Airspeed -65kts**
- **Power -2000RPM**
- **Maintain Altitude and Heading**

Power Off Stall (Stall in Landing configuration)

- **Pre-Maneuver Flow**
- **Power –1500RPM**
- **1st notch below 110kts**
- **2nd & 3rd notch below 85kts**
- **Airspeed 65kts**
- **Descend 200ft**
- **Pitch up for stall**
- **90kts+ return to cruise power setting**

The recovery for both Slow Flight and Power Off Stall are the same. They also happen to be the same procedure for a Go-Around.

Recovery/Go Around

Nose Level & Full Power simultaneously

56kts - Retract 1st notch

74kts with positive rate of climb -2nd & 3rd notches in increments

90 kts+ return to cruise power setting

Steep Turns

- **Pre-maneuver flow**
- **Choose ground reference point**
- **Bank 45 degrees**
- **Add 200RPM**
- **Two Swipes of Nose Up trim**
- **Complete 360degree turn**

Power On Stall (Stall in takeoff configuration)

- **Pre-Maneuver flow**
- **Power back to 1500RPM**
- **Pitch for 55kts**
- **Full Power**
- **Pitch up for Stall**
- **Maintain Coordinated Flight**

Recovery

- **Nose Level**
- **74kts or greater to climb to cruising altitude**

Emergency Engine Failure

A. **Airspeed 68kts**
B. **Best Field**
C. **Checklist**
D. **Declare**

Checkride Guide:

Logbook Entries:

- Dual instruction entries must include instructor name, certificate number, and expiration date
- Log each leg of multi leg flights on separate lines
- Verify all required maneuvers and knowledge areas are recorded
- Log all ground training time and ensure all required subject areas listed
- Check your math
- Total all columns on pages -this can be done in pencil
- Entries must be legible

Endorsements:

- **They must be verbatim from AC 61-65**
- Do not combine endorsements
- Checkride endorsement date should match IACRA endorsement date (this keeps the 60 day and two-month clocks consistent)
- Make sure all endorsements are valid for the applicable dates
- Items to bring:
    - (EXAMINER FEE CASH IS PREFERED)
    - Personal Doc's
        - Photo ID (Government, school, military issues)
        - Student pilot certificate, sport pilot certificate, or recreational pilot certificate
        - Medical certificate (3rd class or better, military flight physical, or basic med)
        - Logbook

Equipment/Publications

- Foggles/view limiting device
- FAR/AIM
- Airplane Flying Handbook
- Pilot's Handbook of Aeronautical Knowledge
- ACS
- EFB or Sectional
- Chart Supplement
- Aircraft POH

Aircraft Records:

- Aircraft maintenance logs (airframe, engine, prop, and AD's)
- Elite Aircraft Audit Sheet for applicable aircraft. To be filled out previously by the student.
- Aircraft airworthiness documents (ARROW)

§61.103 Eligibility Requirements:

- ❑ At least 17 years old §61.103(a)
- ❑ Able to read, speak, write, and understand the English language §61.103(c)

- ❑ Verified US Citizen or TSA authorization for foreign nationals 49 CFR §1552.3(h)
- ❑ Hold a US student pilot certificate, sport pilot certificate, or recreational pilot certificate §61.103(j)
- ❑ Meet aeronautical experience requirements of §61.109 §61.103(g)
- ❑ Receive flight training over areas listed in §61.107(b) §61.103(f)(1)
- ❑ Have passed the required aeronautical knowledge test within the preceding two years §61.103(e)

§61.109 Aeronautical Experience: (TAB all events mentioned below in your logbook)

- ❑ 40 hours flight time §61.109(a)
- ❑ 20 hours dual flight instruction covering the areas of operation listed in §61.107(b) §61.109(a)
- ❑ 3 hours dual cross-country training §61.109(a)(1)
  - ☐ Date of Training: _____
- ❑ 3 hours dual instruction night flight §61.109(a)(2)
- ❑ Dual night cross country over 100 nm miles §61.109(a)(2)(i)
  - ☐ Date of Training :_____ Route Taken: _____
- ❑ 10 takeoffs and landings involving flight in the traffic pattern §61.109(a)(2)(ii)
  - ☐ Date of Training: _____
- ❑ 3 hours instrument dual flight instruction §61.109(a)(3)
- ❑ 3 hours dual flight instruction w/in 2 calendar months §61.109(a)(4)
- ❑ 10 hours solo flight time §61.109(a)(5)
- ❑ 5 hours solo cross country flight time §61.109(a)(5)(i)
- ❑ Date of First Solo Xc: _____
  - ☐ Route Taken:_____
- ❑ Date of Second Solo Xc: _____
  - ☐ Route Taken: _____
- ❑ Date of Third solo Xc: _____
  - ☐ Route Taken: _____
- ❑ Date of Fourth Solo Xc: _____
  - ☐ Route taken: _____
- ❑ Solo cross-country flight, at least 150 nm total distance, full stop landing at a minimum of 3 points, 1 segment of flight consisting of straight-line distance more than 50 nm between takeoff and landing points §61.109(a)(5)(ii)
- ❑ Date of long solo Xc: _____
  - ☐ Route Taken: _____
- ❑ 3 solo takeoffs and landings at an airport with an operating control tower involving flight in the traffic pattern §61.109(a)(5)(iii)
- ❑ Location of towered solo take off/landings:_____
  - ☐ Date completed:_____
- ❑ Record of ground training §61.105(a) Home course of study certificate of completion, or record of ground training from an authorized instructor covering areas of §61.105(b)
- ❑ Instructor logged ground training:_____
- ❑ At home study with certificate of completion:_____ Name of Course:_____
  - ☐ Date of Completion:_____

§61 Required Endorsements:

- ❑ A.14 Endorsement of U.S. citizenship 49 CFR §1552.3(h)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.3 Pre-solo aeronautical knowledge test §61.87(b)(1)
  - ☐ Instructor Initial :_____

- ☐ Student Initials :_____
- ❑ A.4 Pre-solo flight §61.87(c)(1) and (2)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.6 Solo flight first 90 days §61.87(n)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.7 Solo flight each additional 90 days §61.87(p)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.9 Solo cross country flight training §61.93(c)(1) and (2)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.10 Each solo cross country flight §61.93(c)(3)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.32 Aeronautical knowledge test §61.35(a)(1), §61.103(d), §6X105
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.2 Review of deficiencies on the aeronautical knowledge test§61.39(a)(6)(iii)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.1 Prerequisites for practical test §61.39(a)(6)(i) and (ii)
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____
- ❑ A.33 Flight proficiency for the practice test §61.103(f), §61.107(b), §61.109
  - ☐ Instructor Initial :_____
  - ☐ Student Initials :_____

§61 Required Ground:(TAB out in logbook).

- ❑ A. 61.105(b)(1-13) Ground training prior to checkride.
  - ☐ Instructor Initial:_____
  - ☐ Student Initial: _____
- ❑ A. 61.107(b)(1)(i-xii) Ground training prior to first cross country.
  - ☐ Instructor Initial:_____
  - ☐ Student Initial: _____